

Implementing PatchTST: Is a Time Series Worth 42 Words?

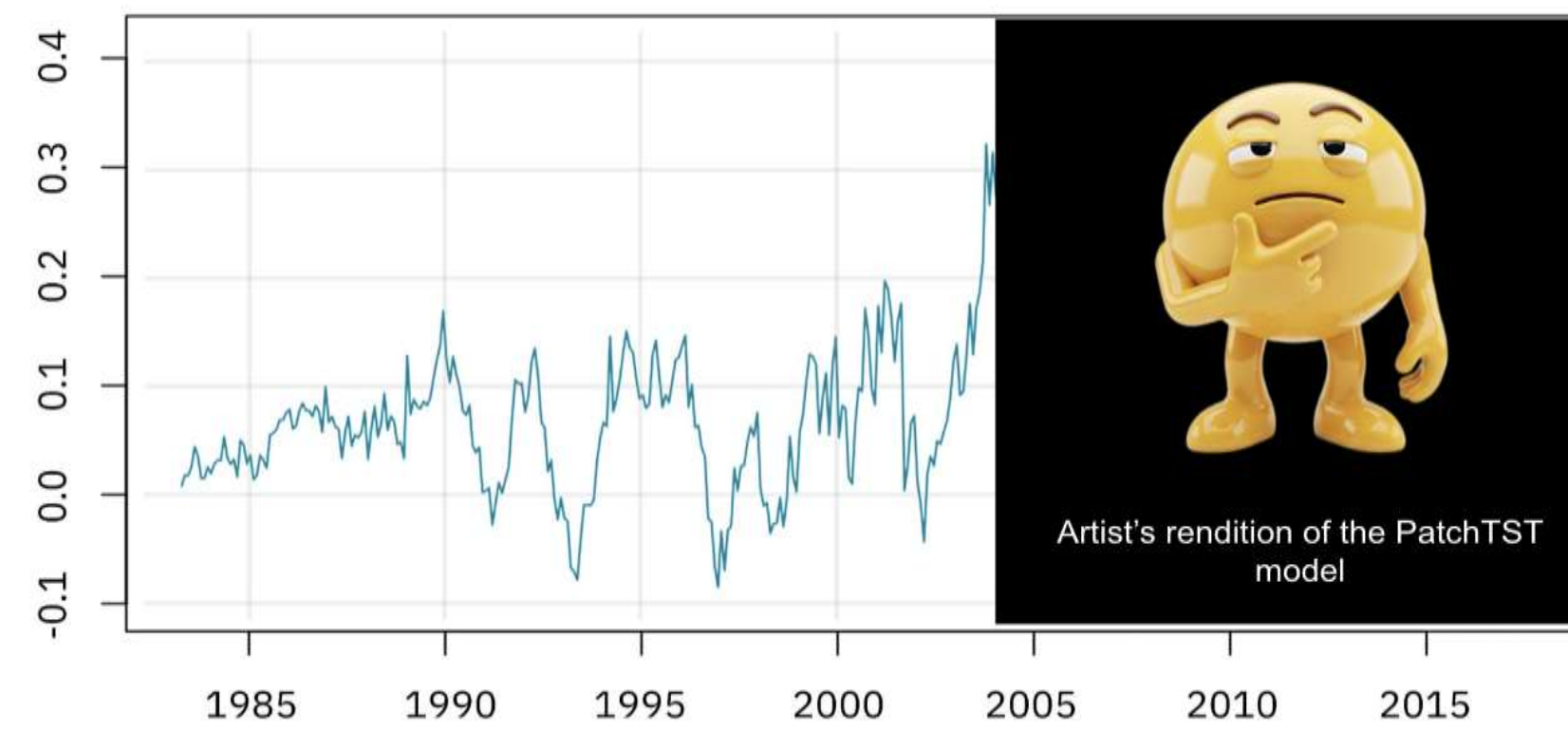
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INTRODUCTION

Transformers suffer from $O(L^2)$ attention complexity over raw timesteps. A simple linear model, DLinear, was proposed by Zeng et al. as proof that transformers were unneeded on time series data. PatchTST (Nie et al., 2023) fixes both issues: time step patching, and channel independence.

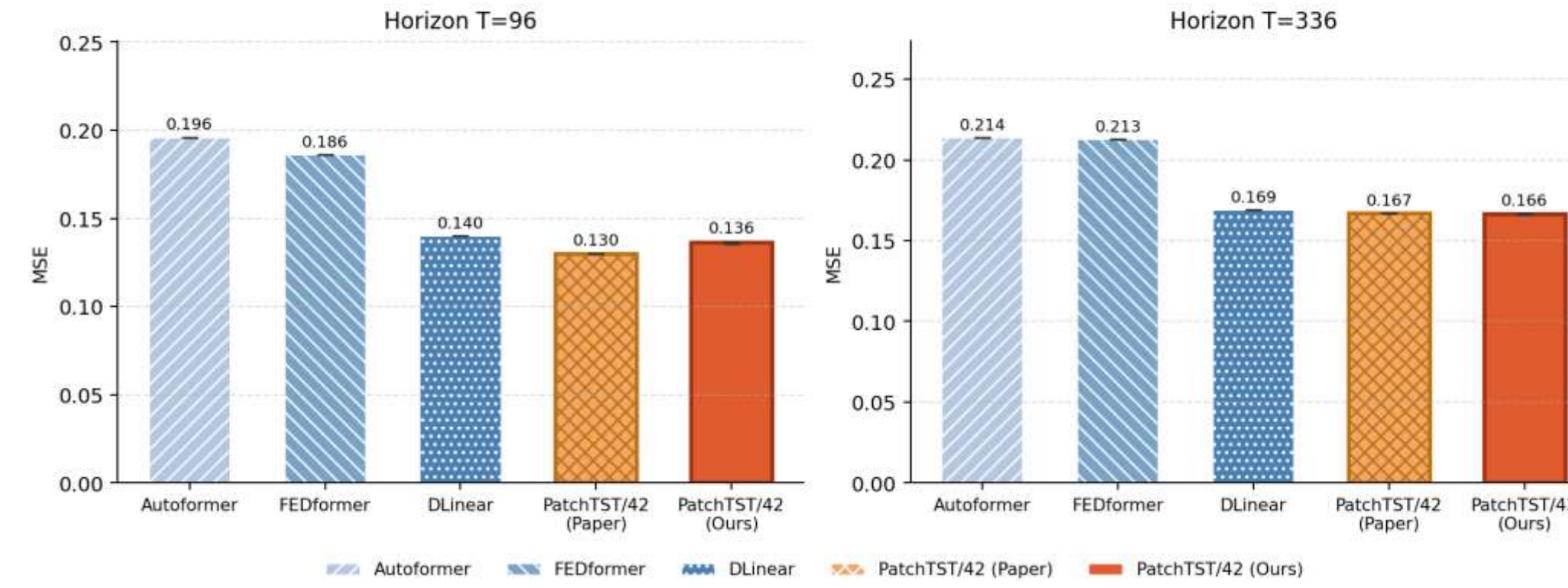
GOAL

- Reproduce PatchTST's supervised forecasting results on Weather and Electricity datasets and test how lookback length, patch shuffling, and channel mixing affect accuracy.

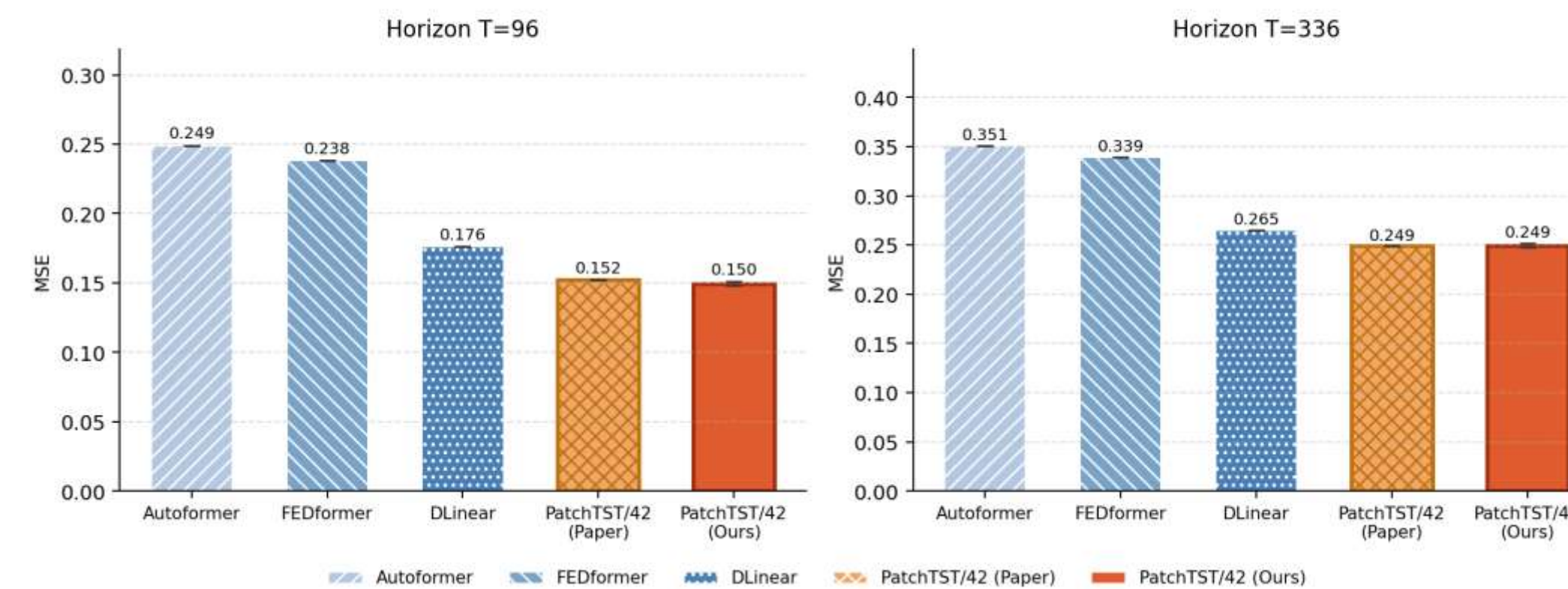


MAIN RESULTS REPLICATION

Electricity Dataset — MSE (lower is better)

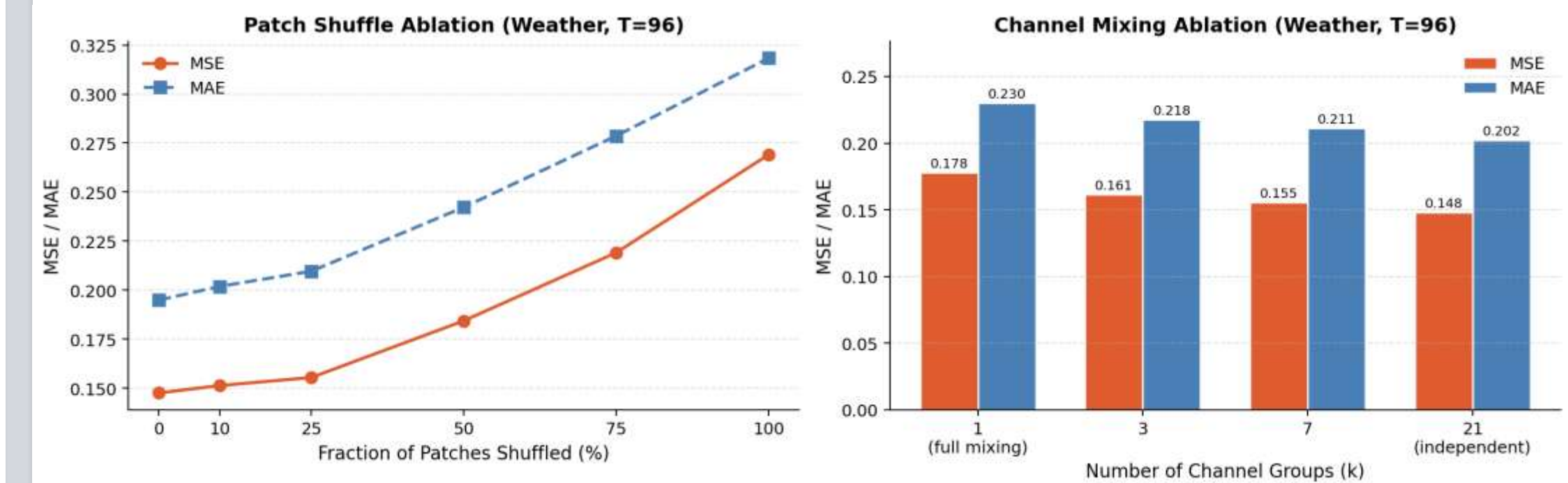


Weather Dataset — MSE (lower is better)



We outperform or match the paper within 0.006 MSE on all. We averaged 3 test runs on each.

INDEPENDENT EXPLORATION



Patch Shuffling

0% → 100% shuffled

MSE 0.148 → 0.269 (+82%)

Channel Mixing

k = 1 → k = 21

MSE 0.148 → 0.178 (+20%)

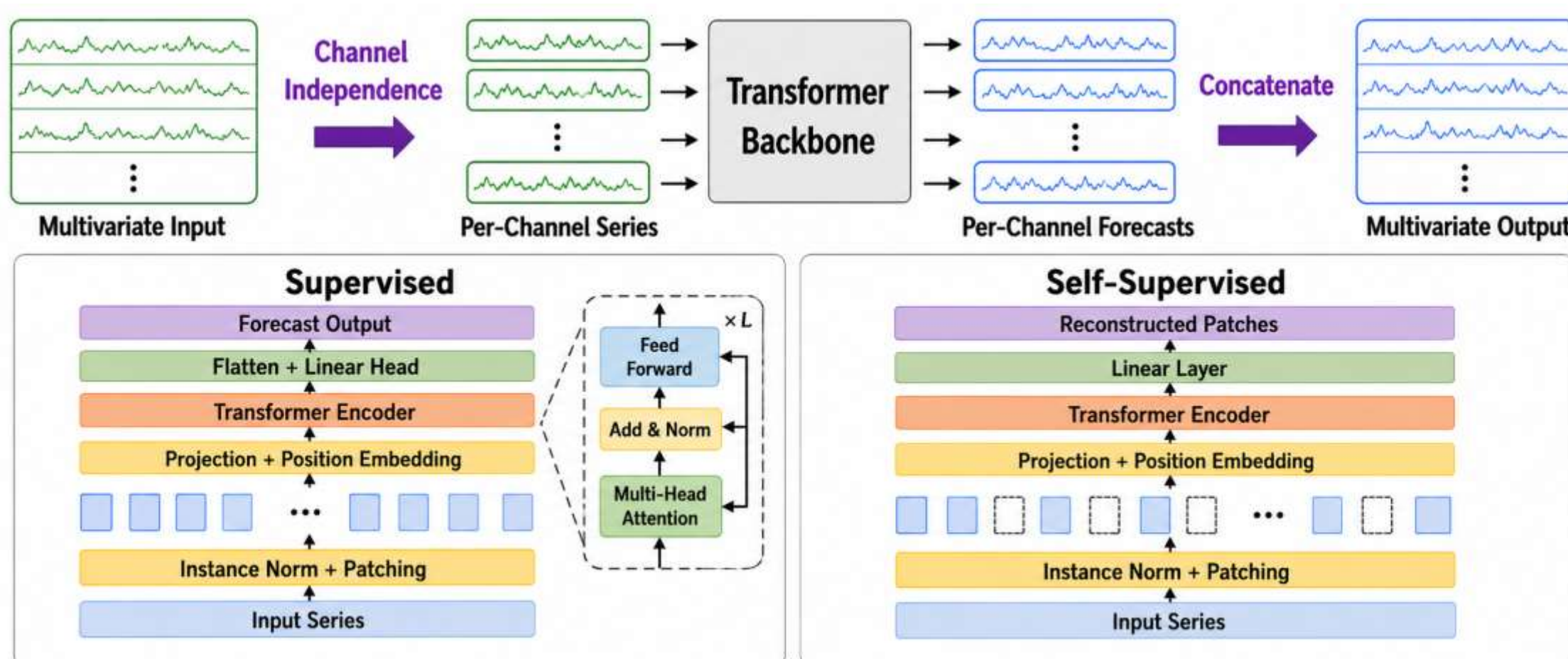
Disrupting patch order or mixing channels both hurt forecasting performance.

METHODOLOGY

We implement PatchTST in PyTorch. This table represents L=336, patch length P=16, stride S=8 with data from Weather with Adam, lr=1e-4, early stopping, and normalized MSE/MAE.

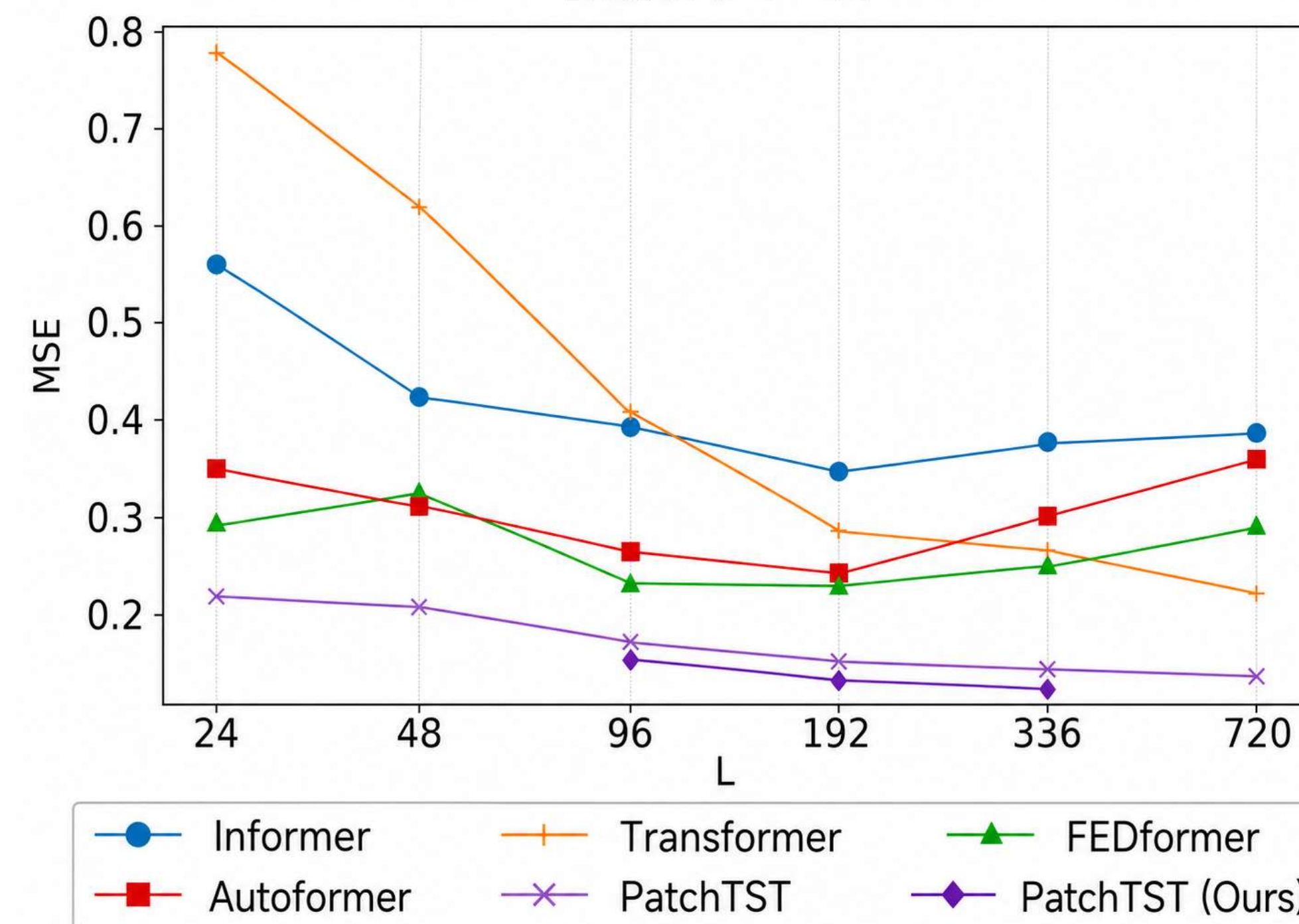
We lacked the compute to replicate SSL, producing 0.156 vs the paper's 0.152 MSE.

Variant	Purpose	Trained?	Params	Tokens N	Attn Pairs	Sec/Epoch
PatchTST/42 (CI)	Main model	Yes	921K	42	1,764	26s
SSL + fine-tune	Tests masked pretraining	Yes	921K	42	1764	30s
No-patch Transformer	Tests patching	Yes	4.58M	337	113,569	431s
Channel-mixing TST	Tests channel indep.	Yes	11.3M	42	1,764	28s
DLinear	Paper-cited baseline	No	—	—	—	—



LOOKBACK WINDOW ABLATION

Weather T=96



Longer lookback windows consistently improve PatchTST, unlike other Transformer baselines.

CONCLUSIONS

- Results fall within 0.006 MSE of the original paper on all four Weather/Electricity settings (T=96, T=336), confirmed across 3 seeds
- Extending lookback from L=96 to L=336 lowers MSE from 0.173 to 0.149 on Weather T=96
- Shuffling patch order raises MSE by up to 82%, evidence that learned temporal structure drives accuracy
- Mixing channels increases MSE by 20%, confirming that independent channel processing is the stronger inductive bias

FUTURE WORK

- Extended self-supervised pretraining, our fine-tuned run (MSE 0.156) suggests gains with larger pretraining budgets
 - Longer prediction horizons (T=720) to evaluate degradation behavior
 - Adaptive patch sizing per channel

REFERENCES

- [1] Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. ICLR 2023.
- [2] Zeng, A., Chen, M., Zhang, L., & Xu, Q. (2023). Are Transformers Effective for Time Series Forecasting? AAAI 2023.
- [3] Wu, H., Xu, J., Wang, J., & Long, M. (2021). Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting. NeurIPS 2021.
- [4] Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L., & Jin, R. (2022). FEDformer: Frequency Enhanced Decomposed Transformer for Long-term Series Forecasting. ICML 2022.